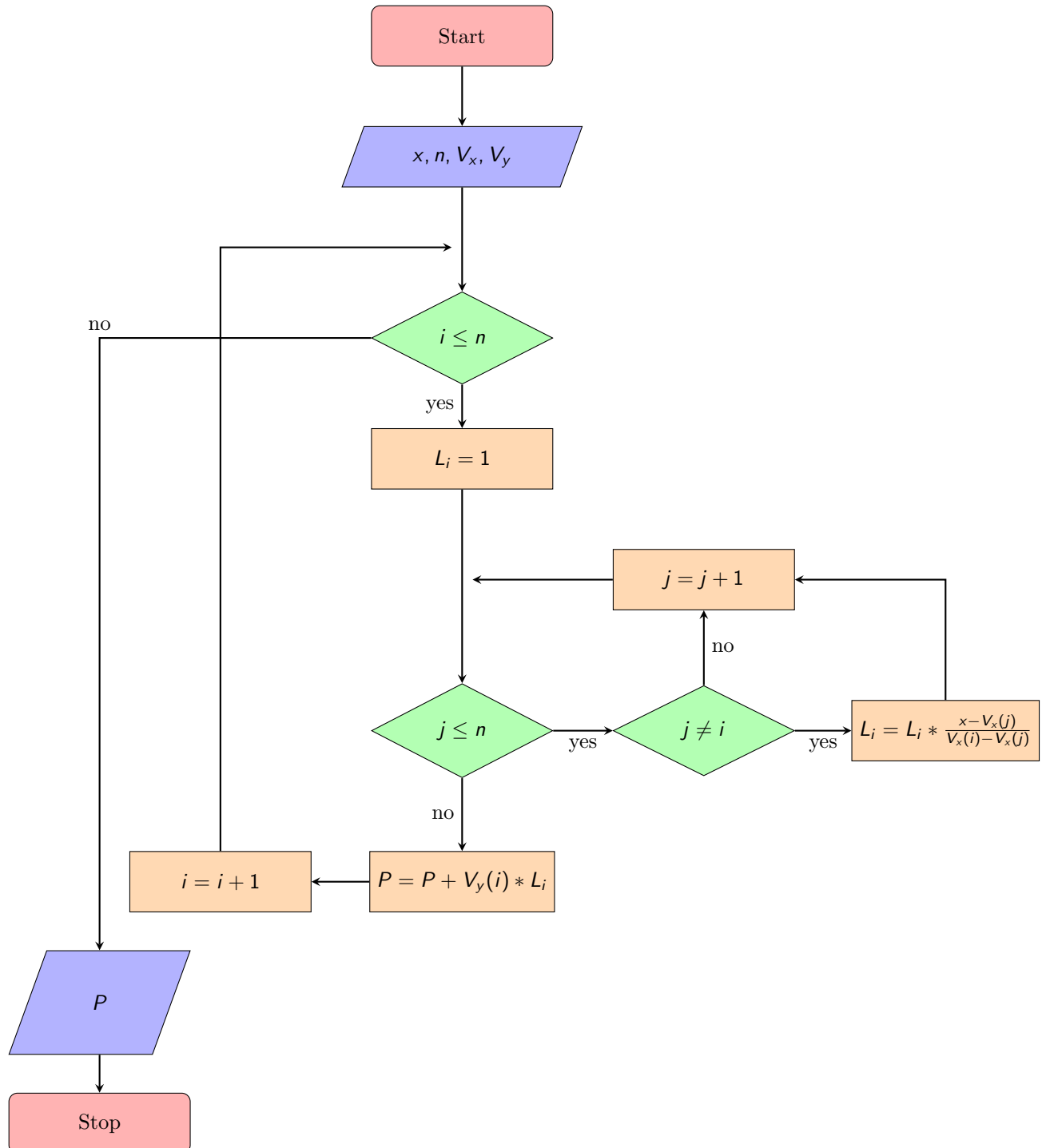


THUẬT TOÁN TÌM ĐA THỨC NỘI SUY LAGRANGE

1 SƠ ĐỒ KHỐI



2 CODE MATLAB

```
1 Vx = [ 0 1.5 2.0 ];
2 Vy = [ 0.000 0.682 0.841 ];
3
4 xp = 1;
5 n = 2;
6
7 P = lagrange( xp, n, Vx, Vy);
8
9 disp(P)
```

```
1 function [ P ] = lagrange( xp, n, Vx, Vy)
2 % Input: xp - la gia tri can tinh f(xp)
3 %       n - bac cua da thuc Lagrange
4 %       Vx - mang gia tri cua x_i
5 %       Vy - mang gia tri cua f(xi)
6 %
7 % Output: P - la gia tri xap xi cua f(xp)
8
9     P = 0;
10
11     for i = 1:n+1
12
13         L = 1;
14         for j = 1:n+1
15
16             if (j ≠ i)
17
18                 L = L*( ( xp-Vx(j) )/( Vx(i)-Vx(j) ) );
19
20             end
21
22         end
23
24         P = P + Vy(i)*L;
25
26     end
27
28 end
```